MATH 341 — Differential Equations

Methods & Techniques Quick Reference

University of Scranton, Mathematics Department

2026-11-04

Table of contents

1 First-Order Equations (Chapter 1) 2
- 1.1 Quick Decision Guide . . . 2
- 1.2 Separation of Variables (§1.2) . . . 3
- 1.3 Integrating Factor / Linear First-Order (§1.3) . . . 3
- 1.4 Exact Equations (§1.4) . . . 4
- 1.5 Substitution Methods (§1.5) . . . 4
  - 1.5.1 Homogeneous Equations . . . 4
  - 1.5.2 Bernoulli Equations . . . 4
  - 1.5.3 Linear Substitution . . . 4
- 1.6 Autonomous Equations & Qualitative Analysis (§1.1, §1.6) . . . 5
- 1.7 Numerical Methods (§1.7) . . . 5
  - 1.7.1 Euler's Method . . . 5
  - 1.7.2 Improved Euler (Heun's Method) . . . 5
  - 1.7.3 Runge-Kutta (RK4) . . . 5

2 Second-Order Linear Equations (Chapter 2) 6
- 2.1 Quick Decision Guide . . . 6
- 2.2 Homogeneous Linear Equations: Characteristic Equation (§2.1) . . . 6
- 2.3 Reduction of Order (§2.2) . . . 6
- 2.4 Undetermined Coefficients (§2.3) . . . 7
  - 2.4.1 Particular Solution Guesses . . . 7
- 2.5 Variation of Parameters (§2.4) . . . 7
- 2.6 Cauchy–Euler Equations (§2.5) . . . 8
- 2.7 Applications: Spring-Mass System (§2.1–§2.3) . . . 8

3 Laplace Transforms (Chapter 3) 9
- 3.1 Definition & Key Properties . . . 9
- 3.2 Common Laplace Transform Table . . . 9
- 3.3 Solving IVPs with Laplace Transforms (§3.1–§3.3) . . . 10
- 3.4 Step Functions & Piecewise Forcing (§3.2) . . . 10
- 3.5 Impulse Functions & Dirac Delta (§3.3) . . . 11
- 3.6 Convolution (§3.4) . . . 11

4 Linear Systems (Chapter 4) 11
4.1 System Setup . . . 11
4.2 Quick Decision Guide: 2×2 Systems . . . 11
4.3 Solving 2×2 Homogeneous Systems (§4.1–§4.2) . . . 12
4.3.1 Case 1: Distinct Real Eigenvalues . . . 12
4.3.2 Case 2: Complex Eigenvalues $\lambda = \alpha \pm \beta i$ . . . 12
4.3.3 Case 3: Repeated Eigenvalue λ (defective) . . . 12
4.4 Phase Plane Analysis (§4.3) . . . 12
4.5 Nonhomogeneous Systems: Variation of Parameters (§4.4) . . . 13

5 Nonlinear Systems & Stability (Chapter 5) 13
5.1 Linearization at Equilibria (§5.1) . . . 13
5.2 Stability Definitions (§5.1–§5.2) . . . 14
5.3 Lyapunov Functions (§5.2) . . . 14
5.4 Limit Cycles & Periodic Orbits (§5.3) . . . 15

6 Overall Strategy Flowchart 15

> **Note**
>
> **Textbook:** Logan, J.D. *A First Course in Differential Equations*, 3rd ed. (Springer, 2015).
> **Sections covered:** All sections except §2.6.
> **Course website:** https://math341diffeq.netlify.app/

---

1 First-Order Equations (Chapter 1)

1.1 Quick Decision Guide

Form of Equation	Technique	See Section
$y' = f(x)\,g(y)$	Separation of Variables	§1.2
$y' + p(x)\,y = q(x)$	Integrating Factor	§1.3
$M\,dx + N\,dy = 0$, $M_y = N_x$	Exact Equation	§1.4
$y' = f(y/x)$	Substitution $v = y/x$	§1.5
$y' = f(ax + by + c)$	Substitution $v = ax + by + c$	§1.5
$y' + p(x)\,y = q(x)\,y^n$	Bernoulli Substitution	§1.5
None of the above	Numerical Methods (§1.6) or Qualitative Analysis	§1.1

1.2 Separation of Variables (§1.2)

When to use: The right-hand side factors as $f(x) \cdot g(y)$.

Form: $\dfrac{dy}{dx} = f(x)\, g(y)$

Steps:

1. Rewrite as $\dfrac{dy}{g(y)} = f(x)\, dx$ (check $g(y) \neq 0$).
2. Integrate both sides: $\displaystyle\int \frac{dy}{g(y)} = \int f(x)\, dx + C$.
3. Solve for y (if possible); apply initial condition to find C.

> **Warning**
>
> **Watch out:** Always check for **equilibrium solutions** where $g(y) = 0$ — these may be lost when dividing.

Example: $y' = xy$
$\Rightarrow \dfrac{dy}{y} = x\, dx \Rightarrow \ln|y| = \dfrac{x^2}{2} + C \Rightarrow y = Ae^{x^2/2}$

1.3 Integrating Factor / Linear First-Order (§1.3)

When to use: Equation is linear in y (or can be put in standard linear form).

Standard form: $y' + p(x)\, y = q(x)$

Steps:

1. Identify $p(x)$ and $q(x)$.
2. Compute integrating factor: $\mu(x) = e^{\int p(x)\, dx}$.
3. Multiply both sides by μ: $(\mu\, y)' = \mu\, q(x)$.
4. Integrate: $\mu\, y = \displaystyle\int \mu\, q(x)\, dx + C$.
5. Solve: $y = \dfrac{1}{\mu}\left(\displaystyle\int \mu\, q\, dx + C\right)$.

> **Tip**
>
> **Tip:** The general solution is $y = y_h + y_p$ where $y_h = C/\mu$ is the homogeneous solution and $y_p = \frac{1}{\mu}\int \mu q\, dx$ is a particular solution.

Example: $y' - 2xy = x$
$p(x) = -2x$, $\mu = e^{-x^2}$, then $(e^{-x^2} y)' = xe^{-x^2}$, giving $y = -\frac{1}{2} + Ce^{x^2}$.

1.4 Exact Equations (§1.4)

When to use: Equation written as $M(x,y)\,dx + N(x,y)\,dy = 0$ with $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$.

Steps:

1. Check exactness: $M_y = N_x$?
2. Find potential function $F(x,y)$ such that $F_x = M$, $F_y = N$.
3. Integrate M w.r.t. x: $F = \displaystyle\int M\,dx + g(y)$.
4. Differentiate F w.r.t. y and set equal to N to find $g'(y)$; integrate.
5. Solution: $F(x,y) = C$.

> **Warning**
>
> **Not exact?** Try integrating factors $\mu(x)$ or $\mu(y)$ to make it exact.
> If $\dfrac{M_y - N_x}{N}$ depends only on x, then $\mu = e^{\int \frac{M_y - N_x}{N}\,dx}$.

1.5 Substitution Methods (§1.5)

1.5.1 Homogeneous Equations

Form: $y' = f\left(\dfrac{y}{x}\right)$
Substitution: $v = y/x$, so $y = vx$ and $y' = v + xv'$.
Reduces to a **separable** equation in v and x.

1.5.2 Bernoulli Equations

Form: $y' + p(x)\,y = q(x)\,y^n$, $n \neq 0, 1$
Substitution: $w = y^{1-n}$, so $w' = (1-n)y^{-n}y'$.
Reduces to a **linear** equation in w.

1.5.3 Linear Substitution

Form: $y' = f(ax + by + c)$
Substitution: $v = ax + by + c$, so $v' = a + by'$.
Reduces to a **separable** equation.

1.6 Autonomous Equations & Qualitative Analysis (§1.1, §1.6)

Form: $y' = f(y)$ (right side depends only on y)

Key concepts:

- **Equilibria (critical points):** Solve $f(y^*) = 0$.
- **Stability:**
  - $f'(y^*) < 0$: **stable** (attractor)
  - $f'(y^*) > 0$: **unstable** (repeller)
- **Phase line:** Plot $f(y)$ vs. y; arrows point right where $f > 0$, left where $f < 0$.

> Tip
>
> **Tip:** The phase line immediately reveals long-term behavior without solving the ODE explicitly.

1.7 Numerical Methods (§1.7)

1.7.1 Euler's Method

Formula: $y_{n+1} = y_n + h\, f(x_n, y_n)$, where h is the step size.

Error: $O(h)$ per step — first-order method. Use small h for accuracy.

1.7.2 Improved Euler (Heun's Method)

Predictor: $\tilde{y}_{n+1} = y_n + h\, f(x_n, y_n)$

Corrector: $y_{n+1} = y_n + \dfrac{h}{2}[f(x_n, y_n) + f(x_{n+1}, \tilde{y}_{n+1})]$

Error: $O(h^2)$ — second-order method.

1.7.3 Runge-Kutta (RK4)

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = f(x_n, y_n), \quad k_2 = f\left(x_n + \tfrac{h}{2}, y_n + \tfrac{h}{2}k_1\right)$$

$$k_3 = f\left(x_n + \tfrac{h}{2}, y_n + \tfrac{h}{2}k_2\right), \quad k_4 = f(x_n + h, y_n + hk_3)$$

Error: $O(h^4)$ — standard workhorse for numerical ODEs.

---

2 Second-Order Linear Equations (Chapter 2)

2.1 Quick Decision Guide

Equation Type	Technique	See Section
$ay'' + by' + cy = 0$	Characteristic Equation	§2.1
$ay'' + by' + cy = g(t)$, $g =$ poly/exp/sin/cos	Undetermined Coefficients	§2.3
$ay'' + by' + cy = g(t)$, $g =$ other	Variation of Parameters	§2.4
$x^2y'' + xy' + \alpha y = 0$	Cauchy–Euler Equation	§2.5
Two solutions y_1, y_2 needed	Reduction of Order	§2.2

2.2 Homogeneous Linear Equations: Characteristic Equation (§2.1)

Standard form: $ay'' + by' + cy = 0$

Characteristic equation: $ar^2 + br + c = 0$, with roots $r = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Discriminant	Roots	General Solution
$b^2 - 4ac > 0$	Two real: $r_1 \neq r_2$	$y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$
$b^2 - 4ac = 0$	Repeated: $r_1 = r_2 = r$	$y = (C_1 + C_2 t)\, e^{rt}$
$b^2 - 4ac < 0$	Complex: $r = \alpha \pm \beta i$	$y = e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$

> **Tip**
>
> For complex roots $r = \alpha \pm \beta i$: $\alpha = -b/(2a)$, $\beta = \sqrt{4ac - b^2}/(2a)$.

2.3 Reduction of Order (§2.2)

When to use: One solution y_1 is known; find a second linearly independent solution y_2.

Substitution: $y_2 = v(t)\, y_1(t)$

Steps:

1. Substitute $y = v\, y_1$ into the ODE.
2. Use $y_1'' + p(t)y_1' + q(t)y_1 = 0$ to eliminate terms.
3. Let $w = v'$ to get a first-order linear ODE in w.

4. Solve for w, then integrate to find v, then $y_2 = v\,y_1$.

Wronskian: $W(y_1, y_2) = y_1 y_2' - y_1' y_2 \neq 0$ confirms linear independence.

Abel's Formula: $W(t) = W(t_0)\, e^{-\int p(t)\,dt}$

2.4 Undetermined Coefficients (§2.3)

When to use: $ay'' + by' + cy = g(t)$ where $g(t)$ is a combination of polynomials, exponentials, sines, or cosines.

General solution: $y = y_h + y_p$

2.4.1 Particular Solution Guesses

Forcing $g(t)$	Guess for y_p
$P_n(t)$ (degree-n poly)	$A_n t^n + \cdots + A_1 t + A_0$
$e^{\alpha t}$	$Ae^{\alpha t}$
$\sin(\beta t)$ or $\cos(\beta t)$	$A\cos(\beta t) + B\sin(\beta t)$
$e^{\alpha t}\sin(\beta t)$ or $e^{\alpha t}\cos(\beta t)$	$e^{\alpha t}(A\cos\beta t + B\sin\beta t)$
$P_n(t)\, e^{\alpha t}$	$(A_n t^n + \cdots + A_0)\, e^{\alpha t}$

> **Warning**
>
> **Modification Rule:** If any term of the guess solves the *homogeneous* equation, multiply the entire guess by t (or t^2 if needed for repeated roots).

2.5 Variation of Parameters (§2.4)

When to use: $y'' + p(t)y' + q(t)y = g(t)$ when $g(t)$ is not of the "undetermined coefficients" type, or the coefficients are non-constant.

Requires: Two known linearly independent solutions y_1, y_2 of the homogeneous equation.

Formulas:

$$y_p = -y_1 \int \frac{y_2\, g}{W}\, dt + y_2 \int \frac{y_1\, g}{W}\, dt$$

where $W = W(y_1, y_2) = y_1 y_2' - y_1' y_2$.

> **Note**
>
> Always write the ODE in **standard form** (leading coefficient 1) before applying this formula.

2.6 Cauchy–Euler Equations (§2.5)

Form: $x^2y'' + \alpha xy' + \beta y = 0$, $x > 0$

Substitution: $x = e^t$ (or try $y = x^r$)

Characteristic equation: $r(r-1) + \alpha r + \beta = 0 \Rightarrow r^2 + (\alpha - 1)r + \beta = 0$

Roots	General Solution
Two real: $r_1 \neq r_2$	$y = C_1 x^{r_1} + C_2 x^{r_2}$
Repeated: $r_1 = r_2 = r$	$y = x^r(C_1 + C_2 \ln x)$
Complex: $r = \lambda \pm \mu i$	$y = x^\lambda(C_1 \cos(\mu \ln x) + C_2 \sin(\mu \ln x))$

2.7 Applications: Spring-Mass System (§2.1–§2.3)

Equation of motion: $my'' + \gamma y' + ky = F(t)$

Parameter	Meaning
$m > 0$	Mass
$\gamma \geq 0$	Damping coefficient
$k > 0$	Spring constant
$F(t)$	External forcing

Damping regimes (unforced, $F = 0$):

Condition	Regime	Behavior
$\gamma^2 - 4mk > 0$	Overdamped	Exponential decay, no oscillation
$\gamma^2 - 4mk = 0$	Critically damped	Fastest decay without oscillation
$\gamma^2 - 4mk < 0$	Underdamped	Oscillates with decaying amplitude
$\gamma = 0$	Undamped	Pure oscillation at $\omega_0 = \sqrt{k/m}$

> ⚠ Warning
>
> **Resonance:** Occurs when $F(t) = F_0 \cos(\omega_0 t)$ and $\gamma = 0$. The amplitude grows without bound — use the **Modification Rule** (multiply guess by t).

---

3 Laplace Transforms (Chapter 3)

3.1 Definition & Key Properties

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt, \quad s > 0$$

Property	Formula
Linearity	$\mathcal{L}\{af + bg\} = aF(s) + bG(s)$
Derivative	$\mathcal{L}\{f'\} = sF(s) - f(0)$
Derivative	$\mathcal{L}\{f''\} = s^2F(s) - sf(0) - f'(0)$
n-th Derivative	$\mathcal{L}\{f^{(n)}\} = s^nF(s) - s^{n-1}f(0) - \cdots - f^{(n-1)}(0)$
s-shifting	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
t-shifting	$\mathcal{L}\{u_c(t)f(t-c)\} = e^{-cs}F(s)$
Convolution	$\mathcal{L}\{f * g\} = F(s)\,G(s)$
Multiplication by t	$\mathcal{L}\{t\,f(t)\} = -F'(s)$

3.2 Common Laplace Transform Table

$f(t)$	$F(s) = \mathcal{L}\{f\}$	Condition
1	$\dfrac{1}{s}$	$s > 0$
t^n	$\dfrac{n!}{s^{n+1}}$	$s > 0$
e^{at}	$\dfrac{1}{s-a}$	$s > a$
$\sin(bt)$	$\dfrac{b}{s^2+b^2}$	$s > 0$
$\cos(bt)$	$\dfrac{s}{s^2+b^2}$	$s > 0$
$e^{at}\sin(bt)$	$\dfrac{b}{(s-a)^2+b^2}$	$s > a$
$e^{at}\cos(bt)$	$\dfrac{s-a}{(s-a)^2+b^2}$	$s > a$

$f(t)$	$F(s) = \mathcal{L}\{f\}$	Condition
$t^n e^{at}$	$\dfrac{n!}{(s-a)^{n+1}}$	$s > a$
$u_c(t)$ (unit step at c)	$\dfrac{e^{-cs}}{s}$	$s > 0$
$\delta(t-c)$ (Dirac delta)	e^{-cs}	

3.3 Solving IVPs with Laplace Transforms (§3.1–§3.3)

When to use: Constant-coefficient ODEs (any order), especially with discontinuous/impulsive forcing.

Steps:

1. Take $\mathcal{L}$ of both sides; use initial conditions automatically.
2. Solve algebraically for $Y(s) = \mathcal{L}\{y(t)\}$.
3. Use **partial fractions** to decompose $Y(s)$.
4. Apply $\mathcal{L}^{-1}$ (use table) to find $y(t)$.

> **Note**
>
> **Partial Fractions Reminder:**
>
> - Distinct linear factors: $\dfrac{A}{s-a} + \dfrac{B}{s-b}$
> - Repeated factor: $\dfrac{A}{s-a} + \dfrac{B}{(s-a)^2}$
> - Irreducible quadratic: $\dfrac{As+B}{s^2+b^2}$ — complete the square if needed

---

3.4 Step Functions & Piecewise Forcing (§3.2)

Heaviside (unit step) function:

$$u_c(t) = \begin{cases} 0, & t < c \\ 1, & t \geq c \end{cases}$$

Writing piecewise functions: Any piecewise function can be written using u_c:

$$f(t) = \begin{cases} g(t), & t < c \\ h(t), & t \geq c \end{cases} = g(t) + [h(t) - g(t)]\, u_c(t)$$

Key formula: $\mathcal{L}\{u_c(t)\, f(t-c)\} = e^{-cs}F(s)$
(shift the function, then multiply by e^{-cs})

3.5 Impulse Functions & Dirac Delta (§3.3)

Dirac delta: $\delta(t-c)$ models an instantaneous impulse at $t = c$.

$$\mathcal{L}\{\delta(t-c)\} = e^{-cs}, \quad c \geq 0$$

$$\int_{-\infty}^{\infty} f(t)\, \delta(t-c)\, dt = f(c)$$

Physical meaning: An impulsive force (hammer blow, lightning strike) applied at time $t = c$.

3.6 Convolution (§3.4)

Definition: $(f * g)(t) = \displaystyle\int_0^t f(t-\tau)\, g(\tau)\, d\tau$

Convolution Theorem: $\mathcal{L}\{f * g\} = F(s)\, G(s)$

When to use: Invert a product $F(s)G(s)$ when partial fractions are unwieldy, or express solutions to IVPs with general forcing as a convolution integral.

---

4 Linear Systems (Chapter 4)

4.1 System Setup

A **first-order linear system:**

$$\mathbf{x}' = A\mathbf{x} + \mathbf{b}(t), \quad \mathbf{x} \in \mathbb{R}^n$$

For a homogeneous system ($\mathbf{b} = \mathbf{0}$), seek solutions $\mathbf{x} = \mathbf{v}\, e^{\lambda t}$, giving the **eigenvalue problem** $A\mathbf{v} = \lambda\mathbf{v}$.

4.2 Quick Decision Guide: 2×2 Systems

Eigenvalues of A	Solution Form	Phase Portrait
Real, distinct, same sign	$C_1\mathbf{v}_1e^{\lambda_1 t} + C_2\mathbf{v}_2e^{\lambda_2 t}$	Node (stable if $\lambda < 0$, unstable if $\lambda > 0$)
Real, distinct, opposite sign	$C_1\mathbf{v}_1e^{\lambda_1 t} + C_2\mathbf{v}_2e^{\lambda_2 t}$	Saddle (always unstable)
Repeated real λ	See below	Improper node
Complex $\lambda = \alpha \pm \beta i$	$e^{\alpha t}(\dots \cos\beta t + \dots \sin\beta t)$	Spiral or center

---

4.3 Solving 2×2 Homogeneous Systems (§4.1–§4.2)

Steps:

1. Find eigenvalues: $\det(A - \lambda I) = 0 \Rightarrow \lambda^2 - \mathrm{tr}(A)\lambda + \det(A) = 0$.
2. For each λ, find eigenvector $\mathbf{v}$: $(A - \lambda I)\mathbf{v} = \mathbf{0}$.
3. Write general solution.

4.3.1 Case 1: Distinct Real Eigenvalues

$$\mathbf{x}(t) = C_1\mathbf{v}_1e^{\lambda_1 t} + C_2\mathbf{v}_2e^{\lambda_2 t}$$

4.3.2 Case 2: Complex Eigenvalues $\lambda = \alpha \pm \beta i$

Compute one complex solution $\mathbf{x}_1 = e^{\alpha t}(\mathbf{p}\cos\beta t - \mathbf{q}\sin\beta t)$ using the real and imaginary parts of the complex eigenvector $\mathbf{v} = \mathbf{p} + i\mathbf{q}$:

$$\mathbf{x}(t) = C_1e^{\alpha t}(\mathbf{p}\cos\beta t - \mathbf{q}\sin\beta t) + C_2e^{\alpha t}(\mathbf{p}\sin\beta t + \mathbf{q}\cos\beta t)$$

4.3.3 Case 3: Repeated Eigenvalue λ (defective)

One eigenvector $\mathbf{v}$; find **generalized eigenvector** $\mathbf{w}$: $(A - \lambda I)\mathbf{w} = \mathbf{v}$.

$$\mathbf{x}(t) = C_1\mathbf{v}\,e^{\lambda t} + C_2(\mathbf{v}\,t + \mathbf{w})\,e^{\lambda t}$$

4.4 Phase Plane Analysis (§4.3)

Trace-Determinant Plane: For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, let $\tau = \mathrm{tr}(A) = a + d$ and $\Delta = \det(A) = ad - bc$.

Region	Eigenvalue Type	Equilibrium
$\Delta < 0$	Real, opposite signs	Saddle (unstable)
$\Delta > 0$, $\tau^2 - 4\Delta > 0$, $\tau < 0$	Real, both negative	Stable node
$\Delta > 0$, $\tau^2 - 4\Delta > 0$, $\tau > 0$	Real, both positive	Unstable node
$\Delta > 0$, $\tau^2 - 4\Delta < 0$, $\tau < 0$	Complex, $\alpha < 0$	Stable spiral
$\Delta > 0$, $\tau^2 - 4\Delta < 0$, $\tau > 0$	Complex, $\alpha > 0$	Unstable spiral
$\Delta > 0$, $\tau = 0$	Pure imaginary	Center (stable)

4.5 Nonhomogeneous Systems: Variation of Parameters (§4.4)

System: $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$

Fundamental matrix: $\Phi(t)$ — columns are linearly independent solutions of $\mathbf{x}' = A\mathbf{x}$.

Particular solution:

$$\mathbf{x}_p(t) = \Phi(t) \int \Phi(t)^{-1}\mathbf{f}(t)\, dt$$

General solution: $\mathbf{x}(t) = \Phi(t)\mathbf{c} + \mathbf{x}_p(t)$

---

5 Nonlinear Systems & Stability (Chapter 5)

5.1 Linearization at Equilibria (§5.1)

System: $\mathbf{x}' = \mathbf{F}(\mathbf{x})$, equilibrium at $\mathbf{x}^* = (x^*, y^*)$ where $\mathbf{F}(\mathbf{x}^*) = \mathbf{0}$.

Jacobian matrix at $\mathbf{x}^*$:

$$J = \begin{pmatrix} \partial F_1/\partial x & \partial F_1/\partial y \\ \partial F_2/\partial x & \partial F_2/\partial y \end{pmatrix}\Bigg|_{\mathbf{x}^*}$$

Linearized system near $\mathbf{x}^*$: $\mathbf{u}' = J\mathbf{u}$ where $\mathbf{u} = \mathbf{x} - \mathbf{x}^*$.

> **Note**
>
> The stability of $\mathbf{x}^*$ for the **nonlinear** system matches the Jacobian classification — **except** when the Jacobian gives a center ($\tau = 0$, $\Delta > 0$): the nonlinear system may be a center or a spiral. Further analysis (e.g., Lyapunov function) is needed.

Steps:

1. Find all equilibria: $F_1(x, y) = 0$ and $F_2(x, y) = 0$ simultaneously.
2. Compute J and evaluate at each equilibrium.

3. Find eigenvalues of J; classify stability.
4. Sketch phase portrait near each equilibrium.

5.2 Stability Definitions (§5.1–§5.2)

Type	Definition
Stable	Nearby trajectories stay nearby
Asymptotically stable	Nearby trajectories stay nearby **and** converge to $\mathbf{x}^*$
Unstable	Trajectories move away from $\mathbf{x}^*$

5.3 Lyapunov Functions (§5.2)

When to use: Proving stability/instability without solving the system; especially useful when linearization is inconclusive (center case).

A function $V(\mathbf{x})$ is a **Lyapunov function** for $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ near $\mathbf{x}^*$ if:

- $V(\mathbf{x}^*) = 0$ and $V(\mathbf{x}) > 0$ for $\mathbf{x} \neq \mathbf{x}^*$ (positive definite).
- $\dot{V} = \nabla V \cdot \mathbf{F} \leq 0$ along trajectories.

Condition on $\dot{V}$	Conclusion
$\dot{V} \leq 0$	$\mathbf{x}^*$ is **stable**
$\dot{V} < 0$	$\mathbf{x}^*$ is **asymptotically stable**
$\dot{V} > 0$ (with $V(\mathbf{x}^*) = 0$)	$\mathbf{x}^*$ is **unstable**

> **Tip**
>
> **Common Lyapunov candidates:** $V = x^2 + y^2$, $V = ax^2 + by^2$, energy functions $H(x, y)$.

5.4 Limit Cycles & Periodic Orbits (§5.3)

Limit cycle: An isolated closed trajectory in the phase plane.

Poincaré–Bendixson Theorem: If a trajectory of $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ in $\mathbb{R}^2$ remains in a bounded region containing no equilibrium, then it must approach a **closed orbit** (limit cycle).

Bendixson's Criterion (no limit cycles): If $\dfrac{\partial F_1}{\partial x} + \dfrac{\partial F_2}{\partial y}$ has constant sign in a simply connected region, then there are **no closed orbits** in that region.

---

6 Overall Strategy Flowchart

```
Given an ODE/System:

  Single ODE, 1st order?
      Separable? → Separation of Variables
      Linear? → Integrating Factor
      Exact? → Exact Equations
      Bernoulli? → Bernoulli Substitution
      Homogeneous (y/x)? → Substitution v = y/x
      Autonomous? → Phase Line / Qualitative Analysis
      None of above → Numerical Methods (Euler / RK4)

  Single ODE, 2nd order, linear, const. coefficients?
      Homogeneous → Characteristic Equation
      Nonhomogeneous?
          g(t) = poly/exp/sin/cos → Undetermined Coefficients
          Other g(t) → Variation of Parameters

  Cauchy-Euler? → Substitution x = e^t

  Use Laplace Transform when:
      Initial conditions given
      Forcing is piecewise (step functions)
      Impulse forcing (Dirac delta)

  System of ODEs?
      Linear, const. coefficients → Eigenvalue Method
          Distinct real → Standard solution
          Complex → Euler's formula
          Repeated → Generalized eigenvectors
      Nonhomogeneous → Variation of Parameters
      Nonlinear → Linearize at equilibria (Jacobian)
          Stability unclear? → Lyapunov Functions
```

Quick Reminders

- Always verify your solution satisfies the original ODE.
- Apply initial/boundary conditions **after** finding the general solution.
- For systems, always check whether the phase portrait classification from the **trace-determinant plane** matches your computed eigenvalues.
- When using Laplace transforms, write the ODE in **standard form** before taking the transform.

MATH 341 — University of Scranton — Fall Semester
Based on Logan, J.D., A First Course in Differential Equations, 3rd ed., Springer, 2015.